

Preparación

docker versión

```
Administrator: Command Prompt

C:\Windows\System32>docker version
Client:
 Version:           27.2.0
 API version:       1.47
 Go version:        go1.21.13
 Git commit:        3ab4256
 Built:             Tue Aug 27 14:17:17 2024
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.34.3 (170107)
Engine:
 Version:          27.2.0
 API version:      1.47 (minimum version 1.24)
 Go version:       go1.21.13
 Git commit:       3ab5c7d
 Built:            Tue Aug 27 14:15:15 2024
 OS/Arch:          linux/amd64
 Experimental:     false
 containerd:
 Version:          1.7.20
 GitCommit:        8fc6bcbff51318944179630522a095cc9dbf9f353
 runc:
 Version:          1.1.13
 GitCommit:        v1.1.13-0-g58aa920
 docker-init:
 Version:          0.19.0
 GitCommit:        de40ad0
```

Docker info

```
Administrator: Command Prompt

C:\Windows\System32>docker info
Client:
 Version:           27.2.0
 Context:           desktop-linux
 Debug Mode:        false
 Plugins:
  buildx: Docker Buildx (Docker Inc.)
    Version: v0.16.2-desktop.1
    Path: C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version: v2.29.2-desktop.2
    Path: C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version: 0.0.24
    Path: C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Alpha) (Docker Inc.)
    Version: v0.0.15
    Path: C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dev: Docker Dev Environments (Docker Inc.)
    Version: v0.1.2
    Path: C:\Program Files\Docker\cli-plugins\docker-dev.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version: v0.2.25
    Path: C:\Program Files\Docker\cli-plugins\docker-extension.exe
  feedback: Provide feedback, right in your terminal! (Docker Inc.)
    Version: v1.0.5
    Path: C:\Program Files\Docker\cli-plugins\docker-feedback.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version: v1.3.0
    Path: C:\Program Files\Docker\cli-plugins\docker-init.exe
  sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
    Version: 0.6.0
    Path: C:\Program Files\Docker\cli-plugins\docker-sbom.exe
  scout: Docker Scout (Docker Inc.)
    Version: v1.13.0
    Path: C:\Program Files\Docker\cli-plugins\docker-scout.exe

Server:
 Containers: 1
  Running: 0
  Paused: 0
  Stopped: 1
 Images: 4
 Server Version: 27.2.0
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
 Cgroup Version: 1
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 nvidia runc
 Default Runtime: runc
 Init Binary: docker-init
 containerd version: 8fc6bcbff51318944179630522a095cc9dbf9f353
```

Docker Image

```
runc version: v1.1.13-0-g58aa920
init version: de40ad0
Security Options:
  seccomp
    Profile: unconfined
Kernel Version: 5.15.153.1-microsoft-standard-WSL2
Operating System: Docker Desktop
OSType: linux
Architecture: x86_64
CPUs: 8
Total Memory: 7.68GiB
Name: docker-desktop
ID: 0bbaf384-0e32-428e-8a96-b51cd9abfc3c
Docker Root Dir: /var/lib/docker
Debug Mode: false
HTTP Proxy: http.docker.internal:3128
HTTPS Proxy: http.docker.internal:3128
No Proxy: hubproxy.docker.internal
Labels:
  com.docker.desktop.address=npipe://\\.\pipe\docker_cli
Experimental: false
Insecure Registries:
  hubproxy.docker.internal:5555
  127.0.0.0/8
Live Restore Enabled: false
```

Descargar una imagen, ver nginx en la lista

```
C:\Windows\System32>docker pull nginx:latest
latest: Pulling from library/nginx
5435b2dcd5c: Pull complete
054715a6bffa: Pull complete
88d1d984b765: Pull complete
4a038fd18db1: Pull complete
84e114c2bb36: Pull complete
7b5d674621c2: Pull complete
448ea5cac5d5: Pull complete
Digest: sha256:7f0adca1fc6c29c8dc49a2e90037a10ba20dc266baaed0988e9fb4d0d8b85ba0
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview nginx:latest

C:\Windows\System32>docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
nginx                latest              a716c9c12c38       13 days ago        161MB
wso2/wso2ei-assenesment-ticxar 6.6.0              a6d191836faa       17 months ago      1.05GB
wso2/wso2ei-telefonica 6.6.0              2ac6f23c0370       17 months ago      7.47GB
<none>              <none>             856a785c44bf       17 months ago      1.05GB
wso2/wso2ei-integrator latest              b798cfdba105       6 years ago        832MB

C:\Windows\System32>
```

Construir imagen Dockerfile

```
File Edit Selection View Go Run Terminal Help
Dockerfile X
C:\Users\User\OneDrive\Escritorio>a>cositas>Betek>docker-lab> Dockerfile
1 FROM alpine:3.20
2 WORKDIR /app
3 RUN echo "Hola desde Docker" > /app/mensaje.txt
4 CMD ["cat", "/app/mensaje.txt"]
```

```
C:\Users\User\OneDrive\Escritorio>a>cositas>Betek>docker-lab>docker build -t myapp:v1 .
[+] Building 5.4s (7/7) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 149B
=> [internal] load metadata for docker.io/library/alpine:3.20
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/3] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
=> => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
=> => sha256:bf8527eb54c3680e728d5b4b383a8ba730d72dae7236fbc8dffa797ed6b224a731 612B / 612B
=> => sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd9f98b50d2e1b96130471 3.63MB / 3.63MB
=> => sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc 9.23kB / 9.23kB
=> => sha256:c64c687cbea9300178b30c95835354e34c4e4febc4badfe27102879de0483b5e 1.02kB / 1.02kB
=> => extracting sha256:25f1d6b1951ac8eb3740558fe94cb83d377bdadf95fd9f98b50d2e1b96130471
=> [2/3] WORKDIR /app
=> [3/3] RUN echo "Hola desde Docker" > /app/mensaje.txt
=> exporting to image
=> => exporting layers
=> => writing image sha256:73b205ad6f89e4ac669200a609e8938d4190514a68c74eeb4c8e4397291b2d1c
=> => naming to docker.io/library/myapp:v1

What's next:
  View a summary of image vulnerabilities and recommendations -> docker scout quickview

C:\Users\User\OneDrive\Escritorio>a>cositas>Betek>docker-lab>
```

```
C:\Users\User\OneDrive\Escritorio>a>cositas>Betek>docker-lab>docker images | findstr myapp
myapp          v1          73b205ad6f89  3 minutes ago  7.81MB
```

```
C:\Users\User\OneDrive\Escritorio>a>cositas>Betek>docker-lab>docker run --rm myapp:v1
Hola desde Docker
```

Etiquetar una imagen

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker tag myapp:v1 myapp:latest

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker images | findstr myapp
myapp          latest       73b205ad6f89 6 minutes ago 7.81MB
myapp          v1          73b205ad6f89 6 minutes ago 7.81MB
```

Inspeccionar una imagen

```
CD Administration Command Prompt
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker image inspect myapp:v1
[
  {
    "Id": "sha256:73b205ad6f89e4ac669200a609e8938d4190514a68c74eeb4c8e4397291b2d1c",
    "RepoTags": [
      "myapp:latest",
      "myapp:v1"
    ],
    "RepoDigests": [],
    "Parent": "",
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-21T02:10:07.872554365Z",
    "DockerVersion": "",
    "Author": "",
    "Config": {
      "Hostname": "",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
      ],
      "Cmd": [
        "cat",
        "/app/mensaje.txt"
      ],
      "ArgsEscaped": true,
      "Image": "",
      "Volumes": null,
      "WorkingDir": "/app",
      "Entrypoint": null,
      "OnBuild": null,
      "Labels": null
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 7807600,
    "GraphDriver": {
      "Data": {
        "LowerDir": "/var/lib/docker/overlay2/kvfyi6cwcqlrph11tfn3f7m3w/diff:/var/lib/docker/overlay2/e86b8e4fa22796d5f7bf6c4417601b62c97dab9b7048f4bdfb28f311c6dfd193/diff",
        "MergedDir": "/var/lib/docker/overlay2/it6yxk6woiulxx7pvkmcxy2em/merged",
        "UpperDir": "/var/lib/docker/overlay2/it6yxk6woiulxx7pvkmcxy2em/diff",
        "WorkDir": "/var/lib/docker/overlay2/it6yxk6woiulxx7pvkmcxy2em/work"
      },
      "Name": "overlay2"
    },
    "RootFS": {
      "Type": "layers",
      "Layers": [
        "sha256:08bc4e534116aa76b16015484b082eac51f9a593416feae9296c8a2d4bb7aa4a2",
        "sha256:f9af93fffd2c9dec54f91fe29b0840bbe87cf4a5b7ee7fb06af568de459182a73",
        "sha256:a13618ad14fb20871363056c61c09bea5466395af42b877af07d81f88b3e3b58"
      ]
    },
    "Metadata": {
      "LastTagTime": "2026-04-21T02:16:01.604125482Z"
    }
  }
]
```

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker image inspect myapp:v1 --format '{{.Id}}'
'sha256:73b205ad6f89e4ac669200a609e8938d4190514a68c74eeb4c8e4397291b2d1c'
```

Guardar una imagen a tar

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker save -o myapp.tar myapp:v1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>dir
Volume in drive C has no label.
Volume Serial Number is B58C-DC80

Directory of C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab

20/04/2026  09:19 p. m.    <DIR>          .
20/04/2026  09:05 p. m.    <DIR>          ..
20/04/2026  09:05 p. m.                112 Dockerfile
20/04/2026  09:19 p. m.      8.112.640 myapp.tar
                2 File(s)      8.112.752 bytes
                2 Dir(s)  116.448.649.216 bytes free

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>dir myapp.tar
Volume in drive C has no label.
Volume Serial Number is B58C-DC80

Directory of C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab

20/04/2026  09:19 p. m.      8.112.640 myapp.tar
                1 File(s)      8.112.640 bytes
                0 Dir(s)  116.447.703.040 bytes free
```

Cargar una imagen desde tar

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker rmi myapp:v1
Untagged: myapp:v1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker images | findstr myapp
myapp          latest       73b205ad6f89   11 minutes ago   7.81MB

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker load -i myapp.tar
Loaded image: myapp:v1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker images | findstr myapp
myapp          latest       73b205ad6f89   12 minutes ago   7.81MB
myapp          v1          73b205ad6f89   12 minutes ago   7.81MB
```

Eliminar una imagen

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker rmi myapp:latest
Untagged: myapp:latest

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker images | findstr myapp
myapp          v1          73b205ad6f89   12 minutes ago   7.81MB
```

Limpiar imágenes no usadas

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker image prune -f
Deleted Images:
deleted: sha256:856a785c44bffa316b2dcf19f4fb145d104803b193def594ef8b1e22e3812e2
deleted: sha256:008e43585339a9deeb2e8b802fe1624a41c737eec183639f63d3748422d2174f

Total reclaimed space: 214.2MB

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>
```

Docker Container

Ejecutar un contenedor

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker run -d --name web1 nginx:latest
420d9ff4fe0367bdbf68913e5471e14975cf7ca96f62a689cd2dc971fba7cc28

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
420d9ff4fe03   nginx:latest  "/docker-entrypoint..."  7 seconds ago  Up 6 seconds  80/tcp       web1
```

Ejecutar contenedor con nombre y puertos

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker run -d --name web2 -p 8080:80 nginx:latest
7f43e507f916a468f7f569f741855fc3bdcdef3d699254e384861f9fd663c8ee
```

 http://localhost:8080

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

Listar todos los contenedores

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS        NAMES
7f43e507f916   nginx:latest  "/docker-entrypoint..."  54 seconds ago  Up 53 seconds  0.0.0.0:8080->80/tcp  web2
420d9ff4fe03   nginx:latest  "/docker-entrypoint..."  2 minutes ago  Up 2 minutes  80/tcp       web1
```

Ver logs

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker logs web2
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/04/21 02:31:00 [notice] 181: using the "epoll" event method
2026/04/21 02:31:00 [notice] 181: nginx/1.29.8
2026/04/21 02:31:00 [notice] 181: built by gcc 14.2.0 (Debian 14.2.0-10)
2026/04/21 02:31:00 [notice] 181: OS: Linux 5.15.153.1, Microsoft Standard WS12
2026/04/21 02:31:00 [notice] 181: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/04/21 02:31:00 [notice] 181: start worker processes
2026/04/21 02:31:00 [notice] 181: start worker process 29
2026/04/21 02:31:00 [notice] 181: start worker process 30
2026/04/21 02:31:00 [notice] 181: start worker process 31
2026/04/21 02:31:00 [notice] 181: start worker process 32
2026/04/21 02:31:00 [notice] 181: start worker process 33
2026/04/21 02:31:00 [notice] 181: start worker process 34
2026/04/21 02:31:00 [notice] 181: start worker process 35
2026/04/21 02:31:00 [notice] 181: start worker process 36
172.17.0.1 - - [21/Apr/2026:02:31:17 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:149.0) Gecko/20100101 Firefox/149.0" "-"
2026/04/21 02:31:18 [error] 30800: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referer: "http://localhost:8080/"
172.17.0.1 - - [21/Apr/2026:02:31:18 +0000] "GET /favicon.ico HTTP/1.1" 404 153 "http://localhost:8080/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:149.0) Gecko/20100101 Firefox/149.0" "-"
```

Inspeccionar un contenedor

```
Administration Command Prompt
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker inspect web2
[
  {
    "Id": "7f43e507f916a468f7f569f741855fc3bdcdef3d699254e384861f9fd663c8ee",
    "Created": "2026-04-21T02:31:05.025004589Z",
    "Path": "/docker-entrypoint.sh",
    "Args": [
      "nginx",
      "-g",
      "daemon off;"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,
      "Pid": 873,
      "Exitcode": 0,
      "Error": "",
      "StartedAt": "2026-04-21T02:31:05.636111999Z",
      "FinishedAt": "0001-01-01T00:00:00Z"
    },
    "Image": "sha256:a716c9c12c382ab51a71127f1dd9440af118939b92af2814d14f6232bb6185d4",
    "ResolvConfPath": "/var/lib/docker/containers/7f43e507f916a468f7f569f741855fc3bdcdef3d699254e384861f9fd663c8ee/resolv.conf",
    "HostnamePath": "/var/lib/docker/containers/7f43e507f916a468f7f569f741855fc3bdcdef3d699254e384861f9fd663c8ee/hostname",
    "HostsPath": "/var/lib/docker/containers/7f43e507f916a468f7f569f741855fc3bdcdef3d699254e384861f9fd663c8ee/hosts",
    "LogPath": "/var/lib/docker/containers/7f43e507f916a468f7f569f741855fc3bdcdef3d699254e384861f9fd663c8ee/7f43e507f916a468f7f569f741855fc3bdcdef3d699254e384861f9fd663c8ee-json.log",
    "Name": "/web2",
    "RestartCount": 0,
    "Driver": "overlay2",
    "Platform": "linux",
    "MountLabel": "",
    "ProcessLabel": "",
    "AppArmorProfile": "",
    "ExecIDs": null,
    "HostConfig": {
      "Binds": null,
      "ContainerIDFile": "",
      "LogConfig": {
        "Type": "json-file",
        "Config": {}
      },
      "NetworkMode": "bridge",
      "PortBindings": {
        "80/tcp": [
          {
            "HostIp": "",
            "HostPort": "8080"
          }
        ]
      },
      "RestartPolicy": {
        "Name": "no",
        "MaximumRetryCount": 0
      },
      "AutoRemove": false,
      "VolumeDriver": "",
      "VolumesFrom": null,
      "ConsoleSize": [
        63,

```

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker inspect web2 --format '{{.State.Status}}'
'running'
```

Detener y arrancar contenedor

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker stop web2
web2

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker start web2
web2

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                NAMES
7f43e507f916   nginx:latest "/docker-entrypoint..." 4 minutes ago  Up 4 seconds  0.0.0.0:8080->80/tcp  web2
420d9ff4fe03   nginx:latest "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  80/tcp               web1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker stop web2
web2

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                NAMES
420d9ff4fe03   nginx:latest "/docker-entrypoint..." 6 minutes ago  Up 6 minutes  80/tcp               web1
```

Entrar en modo interactivo

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker run -it --name alpine1 alpine:3.20 sh
Unable to find image 'alpine:3.20' locally
3.20: Pulling from library/alpine
25f1db01951a: already exists
Digest: sha256:d9e852e87a55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc
Status: Downloaded newer image for alpine:3.20
/ # echo "prueba interactiva"
prueba interactiva
/ # cat /etc/os-release
NAME="Alpine Linux"
ID=alpine
VERSION_ID=3.20.10
PRETTY_NAME="Alpine Linux v3.20"
HOME_URL="https://alpinelinux.org/"
BUG_REPORT_URL="https://gitlab.alpinelinux.org/alpine/aports/-/issues"
/ # exit

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker ps -a | findstr alpine1
8cbc9f2a7011   alpine:3.20          "sh"                  About a minute ago   Exited (0) About a minute ago   alpine1
```

Pausar y reanudar

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker pause web1
web1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker inspect web1 --format '{{.State.Paused}}'
'true'

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker unpause web1
web1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker inspect web1 --format '{{.State.Paused}}'
'false'
```

Eliminar contenedores

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker stop alpine1
alpine1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker rm alpine1
alpine1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker rm -f web1
web1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                NAMES
7f43e507f916   nginx:latest "/docker-entrypoint..." 12 minutes ago  Up 6 minutes  0.0.0.0:8080->80/tcp  web2
```


Volúmenes

Crear volumen

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker volume create datos_nginx
datos_nginx

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker volume ls
DRIVER      VOLUME NAME
local      datos_nginx
```

Inspeccionar volumen

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker volume inspect datos_nginx
[
  {
    "CreatedAt": "2026-04-21T02:47:31Z",
    "Driver": "local",
    "Labels": null,
    "Mountpoint": "/var/lib/docker/volumes/datos_nginx/_data",
    "Name": "datos_nginx",
    "Options": null,
    "Scope": "local"
  }
]
```

Usar volumen en un contenedor

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker run -d --name webvol -p 8081:80 -v datos_nginx:/usr/share/nginx/html nginx:latest
80a56acbe03aa6f0035002da84b33a00ac7676a1007c46205b9a2c6b08ae1184

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>echo "<h1>Hola volumen</h1>" > index.html

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker cp index.html webvol:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to webvol:/usr/share/nginx/html/index.html
```



"

Hola volumen

"

Eliminar volumen

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker rm -f webvol
webvol

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker volume rm datos_nginx
datos_nginx

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker volume ls
DRIVER      VOLUME NAME

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>
```

Redes

Listar redes

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
1ce25efd20fb	bridge	bridge	local
0c5b33b92eb0	host	host	local
343d034e0618	none	null	local

Crear red bridge propia

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker network create my_net
39bb52f45c2aadbd20fac6b9625ad6656484372e45e744d25d1ee719e6015a95

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker network ls
```

NETWORK ID	NAME	DRIVER	SCOPE
1ce25efd20fb	bridge	bridge	local
0c5b33b92eb0	host	host	local
cef0d06cd1e8	mi_red	bridge	local
39bb52f45c2a	my_net	bridge	local
343d034e0618	none	null	local

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>
```

Inspeccionar red

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker network inspect my_net
[
  {
    "Name": "my_net",
    "Id": "39bb52f45c2aadbd20fac6b9625ad6656484372e45e744d25d1ee719e6015a95",
    "Created": "2026-04-21T02:57:50.725115926Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.19.0.0/16",
          "Gateway": "172.19.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Containers": {},
    "Options": {},
    "Labels": {}
  }
]
```

Conectar contenedores a una red

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker run -d --name caja1 --network my_net alpine:3.20 sleep 300
4105d9a0a1c0d5cccede4e84c4de51b528f0c5e306978c00c35f9e49746980bd

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker run -d --name caja2 --network my_net alpine:3.20 sleep 300
b1c3aef716939faeb6e005fd5049fad9074ff807e84add9123b275fa242196ca

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker exec caja1 ping -c 2 caja2
PING caja2 (172.19.0.3): 56 data bytes
64 bytes from 172.19.0.3: seq=0 ttl=64 time=1.101 ms
64 bytes from 172.19.0.3: seq=1 ttl=64 time=0.094 ms

--- caja2 ping statistics ---
2 packets transmitted, 2 packets received, 0% packet loss
round-trip min/avg/max = 0.094/0.597/1.101 ms

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>
```

4.5 Desconectar un contenedor de la red

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker network disconnect my_net caja1

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>docker exec caja1 ping -c 2 caja2
ping: bad address 'caja2'

C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>
```

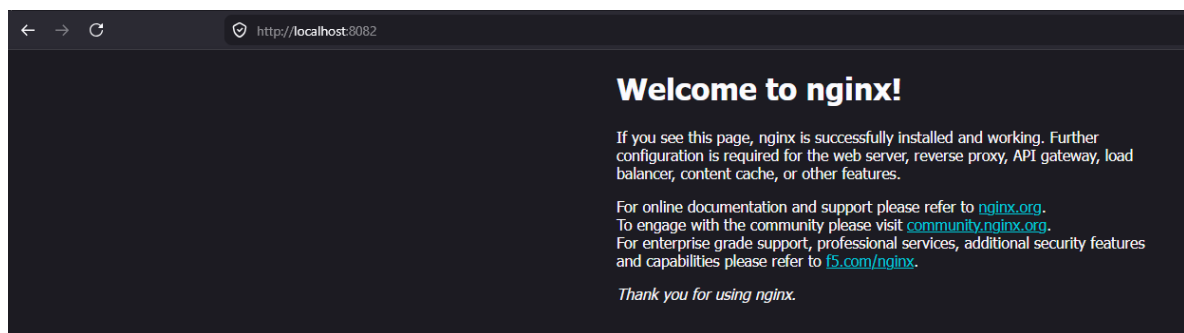
Docker Compose

Crear archivo compose

```
Dockerfile • compose.yml X
C:\Users\User> OneDrive> Escritorio> a> cositas> Betek> docker-lab> compose.yml> {} services> {} web> {} ports
docker-compose.yml - The Compose specification establishes a standard for the definition of multi-container platform-agnostic applications (
1 services:
2   web:
3     image: nginx:latest
4     ports:
5     | - "8082:80"
```

Levantar servicios

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab> docker compose up -d
Running 2/2
Network docker-lab_default created
Container docker-lab-web-1 started
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>
```



Ver logs

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab> docker compose logs
web-1 | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1 | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1 | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1 | 10-listen-on-ipv6-by-default.sh: info: disabled listen on 1000 in /etc/nginx/conf.d/default.conf
web-1 | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1 | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1 | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1 | 2026/04/21 03:08:21 [notice] 1#1: using the "poll" event method
web-1 | 2026/04/21 03:08:21 [notice] 1#1: nginx/1.29.8
web-1 | 2026/04/21 03:08:21 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1 | 2026/04/21 03:08:21 [notice] 1#1: OS: Linux 5.15.153.1-microsoft-standard-WSL2
web-1 | 2026/04/21 03:08:21 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker processes
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker process 29
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker process 30
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker process 31
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker process 32
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker process 33
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker process 34
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker process 35
web-1 | 2026/04/21 03:08:21 [notice] 1#1: start worker process 36
web-1 | 172.20.0.1 - - [21/Apr/2026:03:08:27 +0000] "GET / HTTP/1.1" 200 806 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:149.0) Gecko/20100101 Firefox/149.0" "-"
web-1 | 2026/04/21 03:08:27 [error] 2#829: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.20.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8082", refersr
r: "http://localhost:8082/"
web-1 | 172.20.0.1 - - [21/Apr/2026:03:08:27 +0000] "GET /favicon.ico HTTP/1.1" 404 153 "http://localhost:8082/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:149.0) Gecko/20100101 Firefox/149.0" "-"
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>
```

Bajar servicios

```
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab> docker compose down
Running 2/2
Container docker-lab-web-1 removed
Network docker-lab_default removed
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab> docker compose ps
NAME IMAGE COMMAND SERVICE CREATED STATUS PORTS
C:\Users\User\OneDrive\Escritorio\cositas\Betek\docker-lab>
```

Escalar servicio

```

C:\Users\User\OneDrive\Escritorio\alcosita\Batek\docker-lab>docker compose up -d --scale web=3
[+] Running 3/4
   Network docker-lab_default Created
   Container docker-lab-web-3 Started
   Container docker-lab-web-2 Started
   Container docker-lab-web-1 Created
Error response from daemon: Ports are not available: exposing port TCP 0.0.0.0:8082 -> 0.0.0.0:0: listen tcp 0.0.0.0:8082: bind: Only one user of each socket address (protocol/network address/port) is normally permitted.
C:\Users\User\OneDrive\Escritorio\alcosita\Batek\docker-lab>docker compose ps
NAME                IMAGE              SERVICE      STATUS      PORTS
docker-lab-web-3    nginx:latest      "/docker-entrypoint..."  web         10 seconds ago  Up 9 seconds    0.0.0.0:8082->80/tcp
C:\Users\User\OneDrive\Escritorio\alcosita\Batek\docker-lab>

```

Dockerfile: probar instrucciones principales

Ejemplo corto para probar casi todo

```
C:\Users\User> cd OneDrive\Escritorio> a> cositas> Betek> docker-lab> dockerfile-test> Dockerfile
1 FROM alpine:3.20
2 LABEL autor="prueba"
3 ARG VERSION=1.0
4 ENV APP_ENV=dev
5 WORKDIR /app
6 COPY archivo.txt /app/archivo.txt
7 RUN echo "version=$VERSION" > /app/version.txt
8 ADD archivo.txt /app/archivo2.txt
9 VOLUME ["/data"]
10 CMD ["sh", "-c", "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"]
```

```
C:\Users\User\OneDrive\Escritorio> cd C:\cositas\Betek\docker-lab & mkdir dockerfile-test
```

```
C:\Users\User\OneDrive\Escritorio> cd C:\cositas\Betek\docker-lab & cd dockerfile-test
```

```
C:\Users\User\OneDrive\Escritorio> cd C:\cositas\Betek\docker-lab & dockerfile-test > echo "contenido local" > archivo.txt
```

```
C:\Users\User\OneDrive\Escritorio> cd C:\cositas\Betek\docker-lab & dockerfile-test & dir
Volume in drive C has no label.
Volume Serial Number is 058C-DC38
```

```
Directory of C:\Users\User\OneDrive\Escritorio> cd C:\cositas\Betek\docker-lab & dockerfile-test

20/04/2026 18:16 p. m.    <DIR>          .
20/04/2026 18:16 p. m.    <DIR>          ..
20/04/2026 18:16 p. m.    20 archivo.txt
                        1 file(s)           20 bytes
                        2 dir(s)      111.918.943.213 bytes free
```

```
C:\Users\User\OneDrive\Escritorio> cd C:\cositas\Betek\docker-lab & dockerfile-test & docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 0.2s (1/1) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.15
=> => transferring dockerfile: 351B                             0.18
=> WARN: ConsistentInstructionCasing: Command "/app/version.txt"] should match the case of the command majority (uppercase) (line 11) 0.15
! warning found (use docker --debug to expand):
! ConsistentInstructionCasing: Command "/app/version.txt"] should match the case of the command majority (uppercase) (line 11)
Dockerfile:11
-----
10 |         CMD ["sh", "-c", "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat
11 |     ] >>> /app/version.txt"]
-----
ERROR: failed to solve: dockerfile parse error on line 11: unknown instruction: /app/version.txt"]

C:\Users\User\OneDrive\Escritorio> cd C:\cositas\Betek\docker-lab & dockerfile-test & docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[+] Building 1.6s (18/18) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile              0.05
=> => transferring dockerfile: 351B                             0.10
=> [internal] load metadata for docker.io/library/alpine:3.20   0.06
=> [internal] load dockerimage                                   0.05
=> => transferring context: 2B                                    0.05
=> [1/1] FROM docker.io/library/alpine:3.20                    0.06
=> [internal] load build context                                 0.15
=> => transferring context: 58B                                    0.05
=> CACHED [2/5] WORKDIR /app                                     0.05
=> [3/5] COPY archivo.txt /app/archivo.txt                      0.25
=> [5/5] RUN echo "version:2.0" > /app/version.txt             0.05
=> [5/5] ADD archivo.txt /app/archivo2.txt                     0.25
=> => exporting layers                                           0.25
=> => writing image sha256:e74b09f55723e1dedbc12517ef6aae5bb9a0ac86f6d422e70b23949745db 0.25
=> => naming image shaker.io/library/dockerfile-demo            0.05
```

```
#let's next...
View a summary of image vulnerabilities and recommendations → docker scout quickview

C:\Users\User\OneDrive\Escritorio> cd C:\cositas\Betek\docker-lab & dockerfile-test & docker run --rm dockerfile-demo
APP_ENV=dew
Contenido Local
sh: cat/app/version.txt: not found
```

```
C:\Users\User\OneDrive\Escritorio\acositas\Betek\docker-lab\dockerfile-test>docker build -t dockerfile-demo --build-arg VERSION=2.0 .
[*] Building 0.3s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 352B
=> [internal] load metadata for docker.io/library/alpine:3.20
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/alpine:3.20
=> [internal] load build context
=> => transferring context: 32B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY archivo.txt /app/archivo.txt
=> CACHED [4/5] RUN echo "version=2.0" > /app/version.txt
=> CACHED [5/5] ADD archivo.txt /app/archivo2.txt
=> exporting to image
=> => exporting layers
=> => writing image sha256:a5a40e8267a1bdc4e50945252ae0d52ca76a2f04fb9adc373013d6ccb9afb9c
=> => naming to docker.io/library/dockerfile-demo
```

What's next:

View a summary of image vulnerabilities and recommendations → [docker scout quickview](#)

```
C:\Users\User\OneDrive\Escritorio\acositas\Betek\docker-lab\dockerfile-test>docker run --rm dockerfile-demo
APP_ENV=dev
"contenido local"
version=2.0
```

```
C:\Users\User\OneDrive\Escritorio\acositas\Betek\docker-lab\dockerfile-test>
```

Inspeccionar label, env y volumen

```
C:\Users\User\OneDrive\Escritorio\acositas\Betek\docker-lab\dockerfile-test>docker image inspect dockerfile-demo
[
  {
    "Id": "sha256:a5a40e8267a1bdc4e50945252ae0d52ca76a2f04fb9adc373013d6ccb9afb9c",
    "RepoTags": [
      "dockerfile-demo:latest"
    ],
    "RepoDigests": [],
    "Parent": "",
    "Comment": "buildkit.dockerfile.v0",
    "Created": "2026-04-21T03:17:23.39595755Z",
    "DockerVersion": "",
    "Author": "",
    "Config": {
      "Hostname": "",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "TTY": false,
      "OpenStdin": false,
      "StdinOnce": false,
      "Env": [
        "PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin",
        "APP_ENV=dev"
      ],
      "Cmd": [
        "sh",
        "-c",
        "echo APP_ENV=$APP_ENV && cat /app/archivo.txt && cat /app/version.txt"
      ],
      "ArgsEscaped": true,
      "Image": "",
      "Volumes": {
        "/data": {}
      },
      "WorkingDir": "/app",
      "Entrypoint": null,
      "OnBuild": null,
      "Labels": {
        "autor": "prueba"
      }
    },
    "Architecture": "amd64",
    "Os": "linux",
    "Size": 7887634,
    "GraphDriver": {
      "Data": {
        "lowerDir": "/var/lib/docker/overlay2/15aycgwg7uv37vg84qx1cit0/diff:/var/lib/docker/overlay2/75tj2j1w9j1cidtpq52xoxj/diff:/var/lib/docker/overlay2/kvfyi6cwcqlrph11fn3f7m3w/diff:/var/lib/docker/overlay2/e88b8e4fa22796d5f7b6fc441708b1b2c97d6d0b78a4f8e0f2b311cd0fd19/diff",
        "mergedDir": "/var/lib/docker/overlay2/oyk99kq520upyye2rl1qew3pk/merged",
        "upperDir": "/var/lib/docker/overlay2/oyk99kq520upyye2rl1qew3pk/diff",
        "workDir": "/var/lib/docker/overlay2/oyk99kq520upyye2rl1qew3pk/work"
      },
      "Name": "overlay2"
    },
    "RootFS": {
      "Type": "layers",
      "Layers": [
        "sha256:88bc46534116aa76b16015484b82eac51f9a593416f6ae9296c8a2d4bb7aa42",
        "sha256:f0aef37ff59c0bec4af91fe70b8d8bb37cf4a5b7ae70b69f5606d45937a71"
      ]
    }
  }
]
```

Administrator: Command Prompt

```
C:\Users\User\OneDrive\Escritorio\acositas\Betek\docker-lab\dockerfile-test>docker image inspect dockerfile-demo --format "{{json .Config.Labels}}"
{"autor":"prueba"}

C:\Users\User\OneDrive\Escritorio\acositas\Betek\docker-lab\dockerfile-test>docker image inspect dockerfile-demo --format "{{json .Config.Env}}"
["PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin","APP_ENV=dev"]

C:\Users\User\OneDrive\Escritorio\acositas\Betek\docker-lab\dockerfile-test>docker image inspect dockerfile-demo --format "{{json .Config.Volumes}}"
{"data":{}}
```